Preliminary estimates of extreme Riemann Zeta function peak heights on three lines $s = \{0 + I \cdot t, 1 + I \cdot t, 1.5 + I \cdot t\}$ away from the real axis with $\pm 1\%, \pm 1\%, \pm 0.1\%$ accuracy for $10^{12} < t < 10^{40}$ using spectrally filtered heavily truncated partial Euler Product calculations.

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Executive summary

A comparison of known extreme and large Riemann Siegel Z function peak heights to spectrally filtered heavy truncated $N_{\frac{\sqrt{N_1}}{2}} \sim \frac{1}{2} \cdot \left(\sqrt{\frac{t}{2\pi}}\right)^{0.5}$ partial Euler Products of the Riemann Zeta Z function estimated peak heights on the $s = \{0 + I \cdot t, 1 + I \cdot t\}$ lines with $\pm 1\%$ accuracy for 1e12 < t < 1e21 and the $s = 1.5 + I \cdot t$ line with $\pm 0.1\%$ accuracy for 1e12 < t < 1e40. This behaviour is used to justify publication of the preliminary estimates of the height of extreme or large Riemann Zeta peaks heights (obtained from $N_{\frac{\sqrt{N_1}}{2}}$ based spectrally filtered partial Euler product calculations) on the $s = \{0 + I \cdot t, 1 + I \cdot t\}$ lines in the interval $10^{22} < t < 10^{40}$.

Introduction

Various authors [1-18] have identified large and/or extreme Riemann Siegel Z function peaks **on the critical line**. Of these authors, Gourdon and Sebah [9] and Kotnick [11] presented systematic tabulations of the growth of extreme Riemann Siegel Z function peaks up to t < 10^{13}. Odlyzko and Schönhage [5] and Hiary [12] made significant algorithm improvements in Riemann Zeta function calculations and Tihanyi, Kovacs and Szűcs [15] and Tihanyi, Kovacs and Kovacs [16] introduced improvements to the searching method to find candidate locations for extremely large values of the Riemann zeta function.

In [19], spectrally filtered heavily truncated Euler Products at $N_{\frac{\sqrt{N_1}}{2}} \sim \frac{1}{2} \cdot \left(\sqrt{\frac{t}{2\pi}}\right)^{0.5}$ and $N_{\sqrt{N_1}} \sim \left(\sqrt{\frac{t}{2\pi}}\right)^{0.5}$ rather than N_1 truncation were attempted to provide fast **preliminary** estimates of extreme Riemann Siegel Z function peaks on the critical line in the range $1e8 < t < 1e40$ and exhibited $\pm 10\%$ accuracy for $t > 1e12$ compared to known peak heights.

In practice, spectrally filtered partial Euler Product based Riemann Siegel Z function calculation [19-25] requires a finite interval of the first order partial Euler Product approximation (using truncation at the first quiescent region $N_1 = \sqrt{\frac{t}{2\pi}}$) of the Riemann Siegel Z function to be fourier analyzed, the fourier components are then filtered and scaled conjugate reflection is performed to replaced the dropped frequency components in order to derive a good approximation of the fourier signal of the true Riemann Siegel Z function. The inverse fourier transform of the spectrally filtered partial Euler Product based Riemann Siegel Z function calculation (using truncation at the first quiescent region $N_1 = \sqrt{\frac{t}{2\pi}}$) then represents a good approximation of the true Riemann Siegel Z function for both Riemann Zeta function zero location and lineshape as unnecessary frequency components present in raw partial Euler Product calculations are removed by the fourier analysis with spectral filtering process. A practical issue for spectrally filtered partial Euler Product based Riemann Siegel Z function calculations is that due to the need for a **finite input spectrum interval** for fourier analysis the spectrally filtered partial Euler Product based Riemann Siegel Z function calculation with N_1 truncation of the partial Euler Product is technically slower to execute than the first order Riemann Siegel formula [26-28] which works as a useful approximation for a single t-coordinate away from the real axis. By using spectrally filtered heavily truncated Euler Products at $N_{\frac{\sqrt{N_1}}{2}} \sim \frac{1}{2} \cdot \left(\sqrt{\frac{t}{2\pi}}\right)^{0.5}$ rather than N_1 truncation based on [19] 256 fourier grid points over the interval t=t0+{-6.35,6.4} with spacing $\Delta t = 0.05$ allows faster (than first order Riemann-Siegel formula calculations) preliminary screening of extreme and/or large Riemann Siegel Z function peak heights and location over a wider range of the imaginary axis co-ordinate away from the real axis. (The heavy truncation trades away the precision required for locating Rieman Zeta function zero crossings but has sufficient precision for extreme peak identification.)

In this paper, the use of spectrally filtered partial Euler Product based calculations modified by using heavily truncated Euler Products at $N_{\frac{\sqrt{N_1}}{2}} \sim \frac{1}{2} \cdot \left(\sqrt{\frac{t}{2\pi}}\right)^{0.5}$ are attempted to provide **preliminary** estimates of extreme Riemann Siegel Z function peaks on the $s = \{0 + I \cdot t, 1 + I \cdot t, 1.5 + I \cdot t\}$ lines in the range $1e8 < t < 1e40$ using the approximate functional relationship [25] existing between the negative and positive sidebands of the finite Riemann Siegel Z function fourier transform. Improved accuracy of the spectrally filtered heavily truncated Euler Products away from the critical line is observed since the higher frequency fourier components (which the heavy truncation omits) contribute less to the Riemann Siegel Z function signal away from the critical line.

The input discrete spectra samples, the fourier transforms, the spectral filtering, the zeroth order sideband functional relationship approximation, the rescaled conjugate mirror reflection values and the inverse fourier transforms are calculated using Pari-GP [26] and graphing is performed using R [27] and RStudio IDE [28].

The Riemann Siegel Z function approximately estimated via spectrally filtered partial Euler Products

The Riemann Zeta function is defined [29,32,33], in the complex plane by the integral

$$\zeta(s) = \frac{\prod(-s)}{2\pi i} \int_{C_{\epsilon,\delta}} \frac{(-x)^s}{(e^x - 1)x} dx \tag{1}$$

where $s \,\epsilon\, \mathbb{C}$ and $C_{\epsilon,\delta}$ is the contour about the imaginary poles.

The Riemann Zeta function has been shown to obey the functional equation [29,32,33]

$$\begin{aligned}
\zeta(s) &= \chi(s)\zeta(1-s) && (2)\\
&= 2^s \pi^{s-1} sin(\frac{\pi s}{2})\Gamma(1-s)\zeta(1-s) && (3)\\
&\equiv e^{\log\left[\chi(s)\right]}\zeta(1-s) && (4)\\
&= e^{\left((s-1/2)\cdot\log[\pi] + \log\Gamma\left[(\frac{(1-s)}{2})\right] - \log\left[\Gamma(\frac{s}{2})\right]\right)}\zeta(1-s) && (5)\\
&\equiv e^{-2\theta_{ext}(s)}\zeta(1-s) && (6)
\end{aligned}$$

where

$$\chi(z) = e^{\left((z-1/2)\cdot\log[\pi] + \log\Gamma\left[(\frac{(1-z)}{2})\right] - \log\left[\Gamma(\frac{z}{2})\right]\right)} \tag{7}$$

from equation (5) provides the most numerically stable version of calculating $\chi(z)$ for the Pari-GP [26] calculations performed in this paper.

For $\Re(s) > 1$, the infinite Euler Product of the primes absolutely converges to the infinite Riemann Zeta Dirichlet Series sum [29,32,33]

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s} = \prod_{\rho=2}^{\infty} \frac{1}{(1 - 1/\rho^s)} \qquad \text{for } \Re(s) > 1 \tag{8}$$

Of interest to this paper is obtaining (fast) preliminary estimates of extreme or large Riemann-Siegel Z function peak heights of the Riemann Zeta function across the complex plane at high t

$$Z_\zeta(s) = e^{I \cdot imag(\theta_{ext}(s))} \zeta(s) \tag{9}$$

where (i) equation (8) is used to calculate the $|Z_\zeta(1.5 + I \cdot t)| = |e^{I \cdot imag(\theta_{ext}(1.5+I \cdot t)))} \zeta(s)| = |e^{I \cdot imag(\theta_{ext}(1.5+I \cdot t)))}||\zeta(1.5 + I \cdot t))| \equiv |\zeta(1.5 + I \cdot t)|$ peak height reference values for t > 1e21 used in this paper, and (ii) $\theta ext(s)$ is the extended function version (applicable across the complex plane) of the well known Riemann-Siegel Theta function [29-33] on the critical line given that the spectrally filtered partial Euler Product based approximation of the Riemann-Siegel Z function truncated at the first quiescent region $N_1 = \sqrt{\frac{t}{2\pi}}$ approximated by

$$Z_\zeta(s) \approx e^{I \cdot imag(\theta_{ext}(s))} \left(\prod_{p=2}^{p \leq \lfloor 1.25 \cdot N_1 \rfloor} \frac{1}{(1 - \frac{1}{p^s})} + \chi(s) \cdot \prod_{p=2}^{p \leq \lfloor 1.25 \cdot N_1 \rfloor} \frac{1}{(1 - \frac{1}{p^{(1-(s))}})} \right) \tag{10}$$

$$= EP_1 + EP_2 \tag{11}$$

investigated in [20-25] used fourier analysis of finite spectrum intervals t=(t0-819.15,t0+819.2) based on a 32768 point fourier grid of spacing $\Delta t = 0.05$ was a (more accurate) but slower calculation than first order Riemann Siegel formula calculations [29-31] such as equation (18) below (which requires only a single grid point). The 1.25 hyperparameter is employed [20-25] with full first quiescent region N_1 truncation as it was found to improve decimal place accuracy and its effect is interpreted as helping reduce spectral leakage under fourier analysis.

To improve calculation speed for the **sole** purpose of estimating extreme Riemann Siegel Z function peak heights two improvements were identified [19]

(i) The truncation threshold of the partial Euler Product can be lowered significantly and useful preliminary estimates of the extreme Riemann Siegel Z function peak height are still obtained. In this paper, the results of the following truncation threshold is reported

$N_{\frac{\sqrt{N_1}}{2}} = \frac{1}{2} \cdot \left(\sqrt{\frac{t}{2\pi}} \right)^{0.5}$

$$|Z_\zeta(s)| \approx \left| e^{I \cdot imag(\theta_{ext}(s))} \left(\prod_{p=2}^{p \leq \lfloor N_{\frac{\sqrt{N_1}}{2}} \rfloor} \frac{1}{(1 - \frac{1}{p^s})} + \chi(s) \cdot \prod_{p=2}^{p \leq \lfloor N_{\frac{\sqrt{N_1}}{2}} \rfloor} \frac{1}{(1 - \frac{1}{p^{(1-(s))}})} \right) \right| \tag{12}$$

$$= |EP_1 + EP_2| \tag{13}$$

where when $Re(s) \neq 0.5$ the Riemann Siegel Z function has both real and imaginary components so $|Z_\zeta(s)|$ is the natural way to rank extreme peaks and

(ii) The width of the analyzed spectrum can be lowered to t=(t0-6.35,t0+6.4) based on a 256 point fourier grid of spacing $\Delta t = 0.05$

with minimal loss in the estimated extreme peak height (i.e., the difference in extreme peak height of |Z| > 1000 was $\epsilon < 1$ between using a 32768 point grid and an 256 point grid of the same $\Delta t = 0.05$ spacing and $N_{\frac{\sqrt{N_1}}{2}}$ truncation of the partial Euler Product) resulting in a 128x speed up in estimating extreme Riemann Siegel Z function peak height compared to [20-25] algorithm.

The tradeoff for the above improvement in calculation speed (in preliminary estimates of extreme Riemann Siegel Z function heights) was the loss of precision to identify Riemann Zeta function zero locations.

As Re(s) varies the location of extreme and/or large Riemann siegel Z function peak moves slightly with respect to the critical line location (as well as the magnitude changing). Since this location shift is already covered by the width of the finite fourier sample grid encompassing (t0-6.35,t0+6.4) the nearest grid point to the extreme peak is occasionally found to be shifted by one grid point $\Delta t = 0.05$ with respect its critical line t co-ordinate.

This paper combines the approaches used in [19] (spectral filtering with heavy truncation) and [25] (spectral filtering with conjugate reflection using approximate functional relationship between finite Riemann Siegel Z function sidebands) to present preliminary estimates of extreme and/or large Riemann Siegel Z function peaks for the $s = \{0 + I \cdot t, 1 + I \cdot t, 1.5 + I \cdot t\}$ lines away from the real axis.

Algorithm for spectrally filtered heavily truncated partial Euler Product based Riemann Siegel Z function peak estimates along the $s = \{0 + I \cdot t, 1 + I \cdot t, 1.5 + I \cdot t\}$ lines

The algorithm used for spectral filtering and splicing using Pari-GP language [26] FFT analysis and equation (12) when $Re(s) \neq 0.5$ consisted of

1. Obtain a discrete spectrum sample $\Delta t = (t_0 - 6.35, t_0 + 6.4)$, grid point spacing 0.05, $n \sim 256$ for the term EP_1 (when Re(s) > 0.5) OR the term EP_2 (when Re(s) < 0.5) respectively in equation (12), away from the real axis, using heavy truncation at $N_{\frac{\sqrt{N_1}}{2}}$ to greatly reduce computation time but expecting some inaccuracy in the estimated large and/or extreme peak height.

2. use spectral filtering to retain only the discrete fourier transform components under Pari-GP [26] FFT analysis of (i) EP_1 i.e., the (-,0] angular frequencies OR (ii) EP_2 i.e., the [0,+) angular frequencies

3. splice together either

   the (-,0] angular frequencies from EP_1 and the imputed values by conjugate reflection using the functional sideband relationship [25] prescription (shown in code, where the file "test" is EP_1 and res=Re(s)=0.5,1.5 for this paper)

   ```
   logfun_chi3(res,t)=((res+I*t)-1/2)*log(Pi)+lngamma((1-(res+I*t))/2)-lngamma((res+I*t)/2);
   \\scaling factor for sval > 0.5
   fadj=listcreate(10);
   forstep(k=0,128,1,listput(fadj,abs(exp(logfun_chi3(res+I*exp(k*4*Pi/256/.05)*2*Pi)))));
   k=8;N=2^k;w=FFTinit(k);vhat1=FFT(Vec(test),w);
   fh=(concat((vhat1[1]),concat(conj(Vecrev(vecextract(vhat1,"130..256"))),
   concat((vhat1[129]),vecextract(vhat1,"130..256")))));
   forstep(k=2,128,1,fh[k]=fh[k]*fadj[k]);
   ```

   where vhat1 is the vector output of FFT(EP_1)

   OR

the [0,+) angular frequencies from EP_2 and the imputed values by conjugate reflection using the functional sideband relationship [25] prescription (shown in code, where the file "test" is EP_2 and res=Re(s)=0.0 for this paper)

```
logfun_chi3(res,t)=((res+I*t)-1/2)*log(Pi)+lngamma((1-(res+I*t))/2)-lngamma((res+I*t)/2);
\\scaling factor for sval < 0.5
fadjneg=listcreate(10);
forstep(k=0,128,1,listput(fadjneg,abs(exp(logfun_chi3(1-(res+I*exp(k*4*Pi/256/.05)*2*Pi))))));
k=8;N=2^k;w=FFTinit(k);vhat2=FFT(Vec(test),w);
fh=(concat((vhat2[1]),concat(((vecextract(vhat2,"2..128"))),
concat((vhat2[129]),conj(Vecrev(vecextract(vhat2,"2..128")))))));
forstep(k=2,128,1,fh[256+2-k]=fh[256+2-k]*fadjneg[k]);
```

where vhat2 is the vector output of FFT(EP_2).

4. **OPTIONAL for close spacings between non-trivial zeroes** execute and average fourier analyses of differing lengths from the original stored spectrum. Such averaging was attempted during research for [19] and showed that the **extreme peak height** minimally changed when analyzing 32768, 16384, 8192, 4096, 2048, 1024, 512 and 256 grid points respectively. So the published results only use 256 grid points of grid point spacing $\Delta t = 0.05$. Therefore no averaging results of the fourier analysis of differing lengths was deemed necessary.
5. **OPTIONAL** use interpolation of the final fitted results (spacing 0.05) onto a finer interpolation grid to estimate non-trivial zero positions. The interpolation prescribed this step is not performed in this paper as the single large peak height is the purpose of this paper's calculation.

Note that (i) in the algorithm only one partial Euler Product term needs to be calculated (just EP_1 or EP_2) which reduces the the number of calculations required compared to naive application of equation (12) and (ii) compared to the critical line algorithm described in [19] the restrictions "real(vhat1[1])" and "real(vhat1[129])" appearing in [19] are relaxed to "(vhat1[1])" and "(vhat1[129])" and likewise for their vhat2 counterparts since the DC component of FFT($Z_\zeta(s)$) i.e. element [1] can be complex when $Re(s) \neq 0.5$ and the same convention has been applied to the mid-spectrum element [129].

Results

$\mathbf{s = 0 + I \cdot t}$ line:

Using the above algorithm, Figure 1 presents the log(|Z|) versus log(t) growth

(i) of high precision estimates of peak heights of extreme Riemann Siegel Z function belonging to the locations nearby reported by Kotnick [11] (solid red circles) for t < 1e13 and

(ii) of preliminary estimates of peak heights of extreme Riemann Siegel Z function belonging to the locations nearby reported by (a) Kotnick [11] (black crosses), (b) Hiary [13], Hiary and Bober [14] (blue crosses), (c) Tihanyi et al [16-18] (green crosses) and (d) [19] (magenta triangles) using spectrally filtered heavily truncated Euler Product calculations based on equation (12).

Qualitatively, from figure 1 it can be seen that

- equation (12) estimates of extreme peak height (black crosses) are initially lower than the true peak height for $t < 10^{12}$ (red solid circles).
- the log-log plot of the $s = 0 + I \cdot t$ line extreme peak magnitude growth is **not** exactly linear for the interval shown. As a baseline guide the brown dashed line indicates a naive $\sqrt{\frac{t}{2\pi}}$ growth rate.

The heuristic basis for the illustrated baseline behaviour is that (i) from the functional equation $\zeta(1 + I \cdot t) = \chi(1 + I \cdot t)\zeta(0 + I \cdot t)$, (ii) empirically $1/|\chi(1 + I \cdot t)| \approx \sqrt{\frac{t}{2\pi}}$ for t > 4 for the interval displayed and (iii) presupposing that $|\zeta(1+I\cdot t)|$ for t > 2 has a naive baseline value of 1. Therefore the corresponding naive baseline growth rate for $|\zeta(0+I\cdot t)|$ would be $\frac{1}{\frac{1}{\sqrt{\frac{t}{2\pi}}}} \sim \sqrt{\frac{t}{2\pi}}$. Figure 1 then provides evidence that the extreme peak growth on the $s = 0 + I \cdot t$ line is greater than $\sqrt{\frac{t}{2\pi}}$ in the interval $1e6 < t < 1e40$.

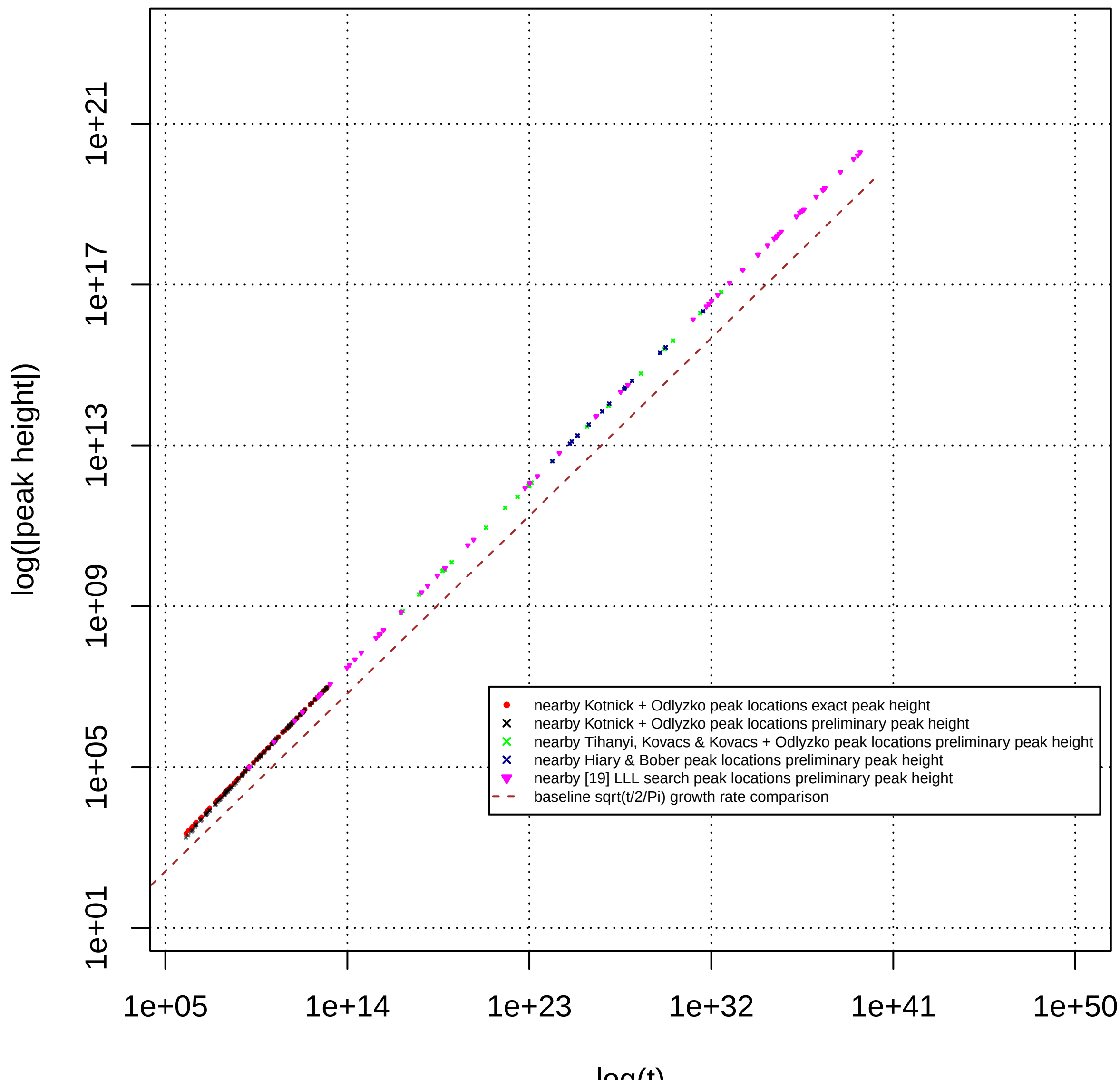

Figure 1. Comparison using log|Z| versus log(t) scatterplot of (i) good precision extreme Riemann Siegel Z function peak heights on the $s = 0 + I \cdot t$ line between 1e6 < t < 1e13 (red solid circles) equation (16) and (ii) preliminary extreme Riemann Siegel Z function peak heights between 1e6 < t < 1e40 based on spectral filtered heavily truncated Euler products based approximation equation (12)

Quantitatively, figure 2 uses the same extreme peak height data from figure 1 but presents the ratios |equation (12)|/|equation (16)| for peaks (i) nearby Kotnick [11] and low lying Tihanyi [16-18] locations (black crosses) and (ii) nearby [19] locations (magenta circles) for t < 1e21.

Hence in figure 2 it can be seen that spectrally filtered heavily truncated estimates using equation (12)

- underestimates the extreme peak height for low t for the $s = 0 + I \cdot t$ line

- there is a slight overshoot the extreme peak height estimate and then evidence of recovery to ratio=1 beyond 1e12. The span of available true peak height data 1e5 < t < 1e21 is however currently **not** sufficient to show whether the approximations based on spectrally filtered equation (12) are properly converging to ratio=1.
- the spread of the ratio values for equation (14) in the region 1e10 < t < 1e21 is roughly (0.99,1.01) so a nominal accuracy performance for spectrally filtered equation (12) estimates is set at ±1% along the $s = 0 + I \cdot t$ line for 1e12 < t < 1e22.
- comparing to figure 2 of [19] the accuracy of the preliminary estimates along the $s = 0 + I \cdot t$ line is much better ±1% than observed along the critical line ±10% which can be explained by the lower contribution of the higher frequency components of the fourier transform of $|Z_\zeta(s)|$ when $Re(s) \neq 0.5$ [25].

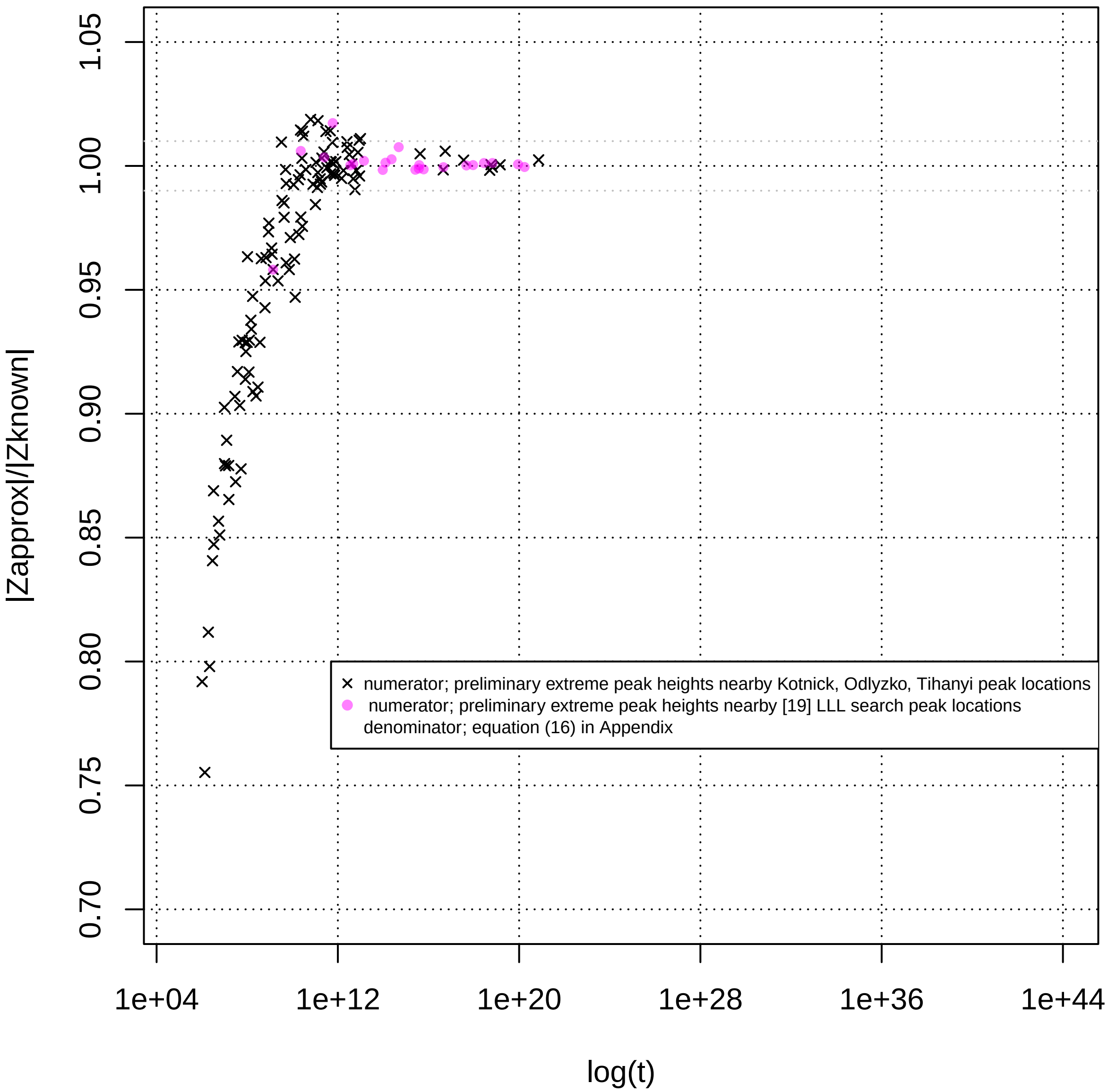

Figure 2. Ratio of spectrally filtered preliminary approximation equation (12) estimates compared to 128 point tapered first order Riemann Siegel formula equation (16) calculations on the $s = 0 + I \cdot t$ line between 1e6 < t < 1e22.

This ±1% accuracy for Re(s)=0 is then speculated to reasonably hold for the wider region 1e10 < t < 1e40 to be tabulated in table 1 given the wider benchmarking that was possible in [19] and figure 9 of this paper for Re(s)=0.5 and Re(s)=1.5 respectively.

Table 1: Grid search location $\Delta t = 0.05$ (column 1) and preliminary height estimate of additional large Riemann Zeta peaks (column 3) using an approximate spectrally filtered Euler Product calculation employing the heavily reduced threshold $N_{\frac{\sqrt{N_1}}{2}} = \frac{1}{2} \cdot \left\lfloor \left(\frac{t}{2\pi}\right)^{0.25} \right\rfloor$. The heavy truncation in the EP calculations results in loss of the precision required for identifying Riemann Zeta zeroes but remains sufficient for preliminary estimates of (~+/- 1% accuracy) large peak heights on the critical line for $1e10 < t < 1e22$ (and by speculation for $1e22 < t < 1e40$). The righthand column (if available) presents the |Z| value obtained using first order Riemann Siegel formula with 128 endpoint tapering calculations.

nearest grid t co-ordinate of peak	|t|	preliminary peak magnitude	|zeta(0.0+I*t)|
2302202919833091938191454510490853528294.00	2.30E39	1944313057103688858883.770 +/-1%	NA
1756509788486806518440112005496184630402.00	1.75E39	1621555145058488203881.936 +/-1%	NA
1083530769932923537813715159168803540949.00	1.08E39	1312664579701170706577.973 +/-1%	NA
244718139402890211698201316945603869614.95	2.44E38	628864386730488315661.904 +/-1%	NA
40980891527423944866379351595135084705.00	4.09E37	248597621308803063393.267 +/-1%	NA
39985965418841204149359159903547304319.00	3.99E37	241997750105506998857.091 +/-1%	NA
32701888308714838788593397151662529169.00	3.27E37	224791787076894586676.960 +/-1%	NA
15305431401376096542184352490753570342.00	1.53E37	152232132650720055507.399 +/-1%	NA
3789770975227925929913752596153535659.00	3.78E36	7313624769720215415.496 +/-1%	NA
3106923501574575498217324013775898531.00	3.10E36	6743557842900054209.048 +/-1%	NA
2383973557258554840502898227029004956.00	2.38E36	6174914510877437573.904 +/-1%	NA
1613337044611093444653528186136557938.00	1.61E36	4905501949589912306.741 +/-1%	NA
284101325235663073414512322415836351.95	2.84E35	2076246465859109758.031 +/-1%	NA
228437899979252082766848904790108956.00	2.28E35	1850999951372771046.597 +/-1%	NA
190153064537019900206349886977800740.00	1.90E35	1658060811248277463.964 +/-1%	NA
167617402698692190502094182200775332.00	1.67E35	1523281443649590374.776 +/-1%	NA
128150196512905705499425650363369432.00	1.28E35	1387213882687408142.204 +/-1%	NA
60758714763756271032716974136770397.00	6.07E34	930536929619052947.883 +/-1%	NA
21230740937947194961712185865535245.00	2.12E34	564799651653303715.182 +/-1%	NA
19885272645274159978060436859288428.00	1.98E34	548743340890013903.751 +/-1%	NA
3598696183652727889399753988496854.95	3.59E33	228512741289707128.144 +/-1%	NA
3546129586355636707276899029799461.00	3.54E33	228847954374815364.887 +/-1%	NA
799009489684392745192175404403618.00	7.99E32	109332264140122742.792 +/-1%	NA
207410624279318332567566187765800.95	2.07E32	54867990864561785.503 +/-1%	NA
103421228622988210350418104528308.00	1.03E32	39081541584168172.473 +/-1%	NA
77636197551578282885077537989878.00	7.76E31	32954078208857492.667 +/-1%	NA
57402689537187327152922167612334.00	5.74E31	28476249722644635.446 +/-1%	NA
12563916677893023747500097076528.00	1.25E31	13529061135766840.094 +/-1%	NA
7536416471667947828910808698.00	7.53E27	317067683587301.512 +/-1%	NA
3286305986924467937604770718.00	3.28E27	212361404750938.347 +/-1%	NA
207977319495059838955685936.00	2.07E26	53671039955057.761 +/-1%	NA
195379681268348910290306782.00	1.95E26	51606692086767.551 +/-1%	NA
3047726497970993730122699.00	3.04E24	6414117433339.727 +/-1%	NA
249774893717336617897828.00	2.49E23	1710083817976.083 +/-1%	NA
103594925224774818957500.00	1.03E23	1140261222687.428 +/-1%	NA
61633023223668798329128.00	6.16E22	858695828946.967 +/-1%	NA
173641788002704982030.00	1.73E20	45164089099.550 +/-1%	45183636245.590
89782960779401678872.95	8.97E19	32664881912.051 +/-1%	32643641931.144
6668512732743895218.00	6.66E18	8830382750.480 +/-1%	8820038884.978
2811690412751902758.90	2.81E18	5703654727.312 +/-1%	5697044785.555
927083404338712008.00	9.27E17	3219323351.441 +/-1%	3218259647.665
472353376312336470.00	4.72E17	2219866408.026 +/-1%	2219452533.668
45698266087910604.00	4.56E16	712702694.460 +/-1%	713070666.111
6107360564274685.00	6.10E15	253913746.543 +/-1%	254266067.259
3897826077577121.00	3.89E15	198811620.863 +/-1%	198752525.496
3758358676276705.00	3.75E15	193810465.306 +/-1%	194013897.355
2641348786316487.00	2.64E15	161943223.012 +/-1%	162192765.024
486153633057302.95	4.86E14	69701365.943 +/-1%	69176714.750
234578622590902.00	2.34E14	47316174.875 +/-1%	47189239.857
126430847573809.95	1.26E14	34129734.866 +/-1%	34085720.150
95913486256898.95	9.59E13	29572095.440 +/-1%	29619932.479
14223012917203.00	1.42E13	11475603.915 +/-1%	11452031.799
4779391590262.05	4.77E12	6455719.203 +/-1%	6450055.076
3556019595338.05	3.55E12	5544234.143 +/-1%	5542856.436
594979079066.05	5.94E11	2286244.573 +/-1%	2247410.031
249284579994.95	2.49E11	1403852.026 +/-1%	1399023.121
23065530804.95	2.30E10	422578.873 +/-1%	420040.253
1387123310.00	1.38E9	99184.210 +/-1%	103517.944
102805259.025	1.02E8	24822.570 +/-1%	25766.801

Table 1 presents a comparison of the preliminary estimates using equation (12) and the 3+ decimal place accuracy of equation (16) estimates for absolute value extreme Riemann Siegel Z function peaks $|Z_\zeta(0 + I \cdot t)|$ along the $s = 0 + I \cdot t$ line. The peak locations are nearby the peak locations reported by [19] on the critical line. Typically, given the results are from a fourier analysis grid with input spectrum spacing $\Delta t = 0.05$ the peak locations along the $s = 0 + I \cdot t$ line are only $\{-0.05, 0, 0.05\}$ different from the locations reported in [19] observed along the critical line.

Of particular note is that the growth of $|Z_\zeta(0 + I \cdot t)|$ for these extreme peak locations in Table 1 appears strongly monotonic which was not the case for the corresponding table in [19] on the critical line.

$\mathbf{s = 1 + I \cdot t}$ line:

Using the above spectral filtering algorithm and equation (16), Figure 3 presents the |Z| versus log(t) growth of

(i) good precision estimates of peak heights of extreme Riemann Siegel Z function belonging to the locations nearby reported by Kotnick [11] (solid red circles) for t < 1e13 based on equation (16) and

(ii) preliminary estimates of peak heights of extreme Riemann Siegel Z function belonging to the locations nearby reported by (a) Kotnick [11] (black crosses), (b) Hiary [13], Hiary and Bober [14] (blue crosses), (c) Tihanyi et al [16-18] (green crosses) and (d) [19] (magenta triangles) using spectrally filtered heavily truncated Euler Product calculations based on equation (12).

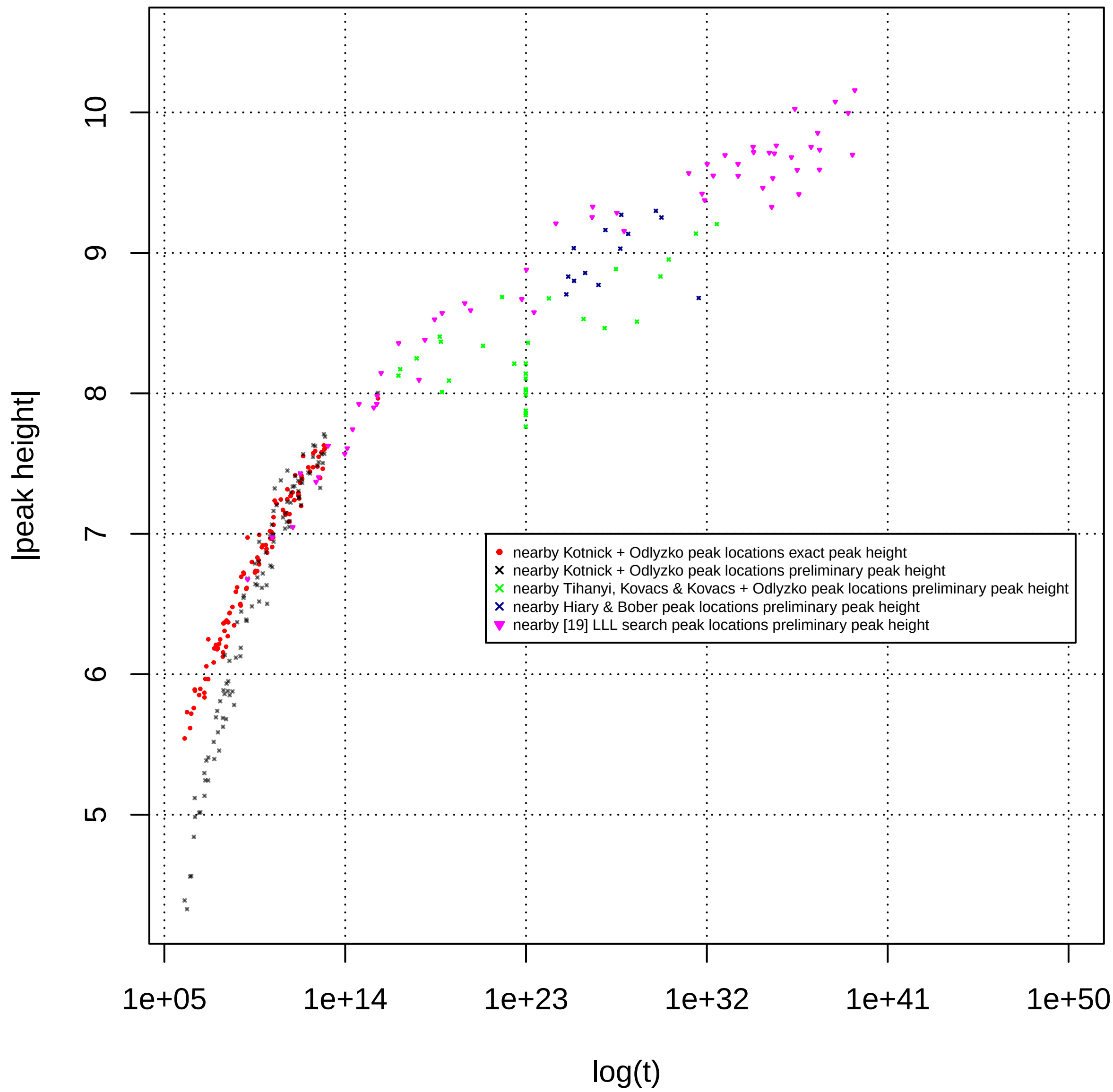

Figure 3. Comparison using |Z| versus log(t) scatterplot of (i) good precision extreme Riemann Siegel Z function peak heights on the $s = 1 + I \cdot t$ line between 1e6 < t < 1e13 (red solid circles) equation (16) and (ii) preliminary extreme Riemann Siegel Z function peak heights between 1e6 < t < 1e40 based on spectral filtered heavily truncated Euler products based approximation equation (12)

Qualitatively, from figure 3 it can be seen that

- equation (12) estimates of extreme peak height (black crosses) are initially lower than the true peak height for $t < 10^9$ (red solid circles) but eventually reach reasonable agreement with the good precision equation (16) estimates between $10^9 < t < 10^{13}$.

Quantitatively, figure 4 uses the same extreme peak height data from figure 3 but presents the ratios |equation (12)|/|equation (16)| for peaks (i) derived nearby Kotnick [11] and low lying Tihanyi [16-18] locations (black crosses) and (ii) nearby [19] locations (magenta circles) for t < 1e21.

Hence in figure 4 it can be seen that spectrally filtered heavily truncated estimates using equation (12)

- underestimates the extreme peak height for low t for the $s = 1 + I \cdot t$ line
- there is a slight overshoot the extreme peak height estimate and then evidence of recovery to ratio=1 beyond 1e12. The span of available true peak height data 1e5 < t < 1e21 is however currently **not** sufficient to show whether the approximations based on spectrally filtered equation (12) are properly converging to ratio=1.
- the spread of the ratio values for equation (12) in the region 1e10 < t < 1e21 is roughly (0.99,1.01) so a nominal accuracy performance for spectrally filtered equation (12) estimates is set at ±1% along the $s = 0 + I \cdot t$ line for 1e12 < t < 1e22.
- comparing to figure 2 of [19] the accuracy of the preliminary estimates along the $s = 1 + I \cdot t$ line is much better ±1% than observed along the critical line ±10% which can be explained by the lower contribution of the higher frequency components of the fourier transform of $|Z_\zeta(s)|$ when $Re(s) \neq 0.5$ [25].

On even cursory inspection figures 2 and 4 are basically the same scatterplots but independently calculated using $s = 0 + I \cdot t$ and $s = 1 + I \cdot t$ inputs and in hindsight should exhibit the same ratio behaviour due to the Riemann Zeta functional equation.

Numerically, the independent spectral filtering based calculations using 58 digit precision under Pari-GP [26] computation do closely follow the functional equation behaviour, for example

when t0=927083404338712008,

At $s = 0 + I \cdot t0$, |Z|=3219323351.4411029188415423827222407873067510328955 using spectral filtering of equation (12)

using the functional equation $|Z| \cdot \chi(1 + I \cdot t0) \equiv 8.3809862956570668036325560070816853423454059289737$

At $s = 1 + I \cdot t0$, |Z|=8.3809862956570668036270338532171408878901737023153 using spectral filtering of equation (12)

So the Pari-GP [26] based spectral filtering calculations contain enough precision to demonstrate the Riemann Zeta functional equation behaviour.

Currently, figure 4 contains three additional data points (compared to figure 2) by accessing the CoCalc platform [34] at

(1) $s = 1 + I \cdot 6436526919750171929565.996$ i.e. 6.43e21

|Zknown|=8.6834122929229958833373942001201198419876975974308 equation (16) using CoCalc platform [34]

CPU time=4x47.5 = 190 hrs

|Zapprox|=8.6856918716480791787548727571430305415832130555602 spectrally filtered equation (12)

CPU time=6 secs

ratio |Zapprox|/|Zknown| = 1.000262...

(2) $s = 1 + I \cdot 25911478989738226351131.25$ i.e. 2.59e22

|Zknown|=8.2070918810250663115169424329539519919990656494285 equation (16) using CoCalc platform [34]

CPU time=4x95.4 = 381.6 hrs

|Zapprox|=8.2113526381385702225381909590260043107136931326172 spectrally filtered equation (12)

CPU time=7 secs

ratio |Zapprox|/|Zknown| = 1.000519...

(3) $s = 1 + I \cdot 61633023223668798329128$ i.e. 6.16e22

|Zknown|=8.6713697412080909095144095456390531217979325211402 equation (16) using CoCalc platform [34]

CPU time=4x160.5 = 642 hrs

|Zapprox|=8.6700689622340724331396260195552563080921009037858 spectrally filtered equation (12)

CPU time=8.5 secs

ratio |Zapprox|/|Zknown| = 0.999849...

as further benchmarking testing continues. It should be noted that equation (16) has low RAM requirements but the spectral filtered equation (12) calculations have increasing RAM requirements as t increases in order to reuse log(prime) values for all 256 grid points (hence why the current published t value limit < 1e40 in this paper).

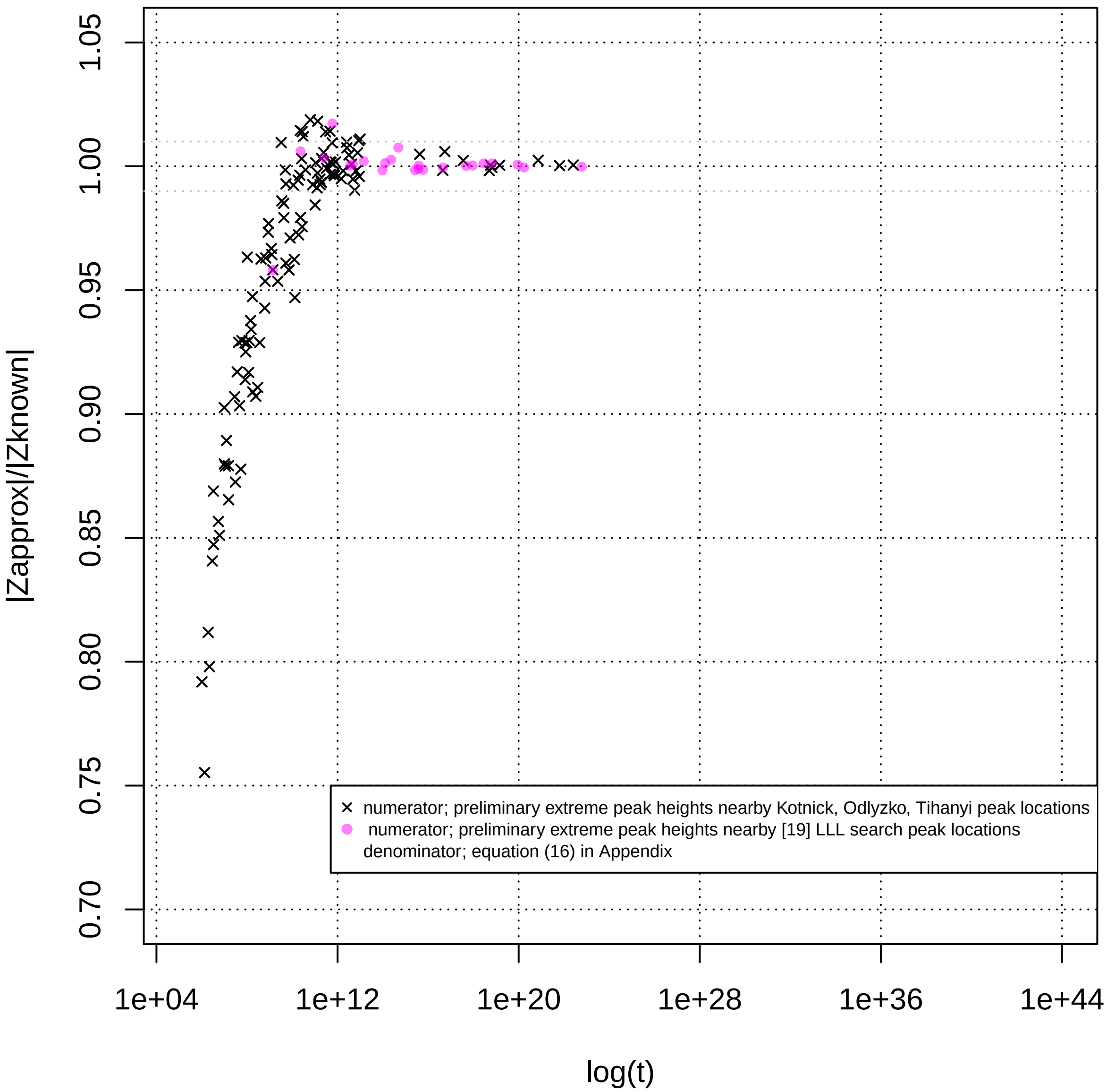

Figure 4. Ratio of spectrally filtered preliminary approximation equation (12) estimates compared to 128 point tapered first order Riemann Siegel formula equation (16) calculations on the $s = 1 + I \cdot t$ line between 1e6 < t < 1e22.

This $\pm 1\%$ accuracy for Re(s)=1, in figure 4 is then speculated to reasonably hold for the wider region 1e10 < t < 1e40 to be tabulated in table 2 given the wider benchmarking that was possible in [19] and figure 9 of this paper for Re(s)=0.5 and Re(s)=1.5 respectively.

Table 2: Grid search location $\Delta t = 0.05$ (column 1) and preliminary height estimate of additional large Riemann Zeta peaks (column 3) using an approximate spectrally filtered Euler Product calculation employing the heavily reduced threshold $N_{\frac{\sqrt{N_1}}{2}} = \frac{1}{2} \cdot \left\lfloor \left(\frac{t}{2\pi}\right)^{0.25} \right\rfloor$. The heavy truncation in the EP calculations results in loss of the precision required for identifying Riemann Zeta zeroes but remains sufficient for preliminary estimates of (~+/- 1% accuracy) large peak heights on the critical line for $1e10 < t < 1e22$ (and by speculation for $1e22 < t < 1e40$). The righthand column (if available) presents the |Z| value obtained using first order Riemann Siegel formula with 128 endpoint tapering calculations.

nearest grid t co-ordinate of peak	|t|	preliminary peak magnitude	|zeta(1+I*t)|
2302202919833091938191454510490853528294.00	2.30E39	10.157441 +/-1%	NA
1756509788486806518440112005496184630402.00	1.75E39	9.698316 +/-1%	NA
1083530769932923537813715159168803540949.00	1.08E39	9.995986 +/-1%	NA
244718139402890211698201316945603869614.95	2.44E38	10.076596 +/-1%	NA
40980891527423944866379351595135084705.00	4.09E37	9.734108 +/-1%	NA
39985965418841204149359159903547304319.00	3.99E37	9.592845 +/-1%	NA
32701888308714838788593397151662529169.00	3.27E37	9.853351 +/-1%	NA
15305431401376096542184352490753570342.00	1.53E37	9.753791 +/-1%	NA
3789770975227925929913752596153535659.00	3.78E36	9.417077 +/-1%	NA
3106923501574575498217324013775898531.00	3.10E36	9.589892 +/-1%	NA
2383973557258554840502898227029004956.00	2.38E36	10.024671 +/-1%	NA
1613337044611093444653528186136557938.00	1.61E36	9.680790 +/-1%	NA
284101325235663073414512322415836351.95	2.84E35	9.764099 +/-1%	NA
228437899979252082766848904790108956.00	2.28E35	9.707608 +/-1%	NA
190153064537019900206349886977800740.00	1.90E35	9.531005 +/-1%	NA
167617402698692190502094182200775332.00	1.67E35	9.326324 +/-1%	NA
128150196512905705499425650363369432.00	1.28E35	9.713454 +/-1%	NA
60758714763756271032716974136770397.00	6.07E34	9.462791 +/-1%	NA
21230740937947194961712185865535245.00	2.12E34	9.716319 +/-1%	NA
19885272645274159978060436859288428.00	1.98E34	9.754239 +/-1%	NA
3598696183652727889399753988496854.95	3.59E33	9.548337 +/-1%	NA
3546129586355636707276899029799461.00	3.54E33	9.632958 +/-1%	NA
799009489684392745192175404403618.00	7.99E32	9.695323 +/-1%	NA
207410624279318332567566187765800.95	2.07E32	9.549782 +/-1%	NA
103421228622988210350418104528308.00	1.03E32	9.631874 +/-1%	NA
77636197551578282885077537989878.00	7.76E31	9.374907 +/-1%	NA
57402689537187327152922167612334.00	5.74E31	9.421208 +/-1%	NA
12563916677893023747500097076528.00	1.25E31	9.567425 +/-1%	NA
7536416471667947828910808698.00	7.53E27	9.155023 +/-1%	NA
3286305986924467937604770718.00	3.28E27	9.285633 +/-1%	NA
207977319495059838955685936.00	2.07E26	9.328717 +/-1%	NA
195379681268348910290306782.00	1.95E26	9.254570 +/-1%	NA
3047726497970993730122699.00	3.04E24	9.209559 +/-1%	NA
249774893717336617897828.00	2.49E23	8.576951 +/-1%	NA
103594925224774818957500.00	1.03E23	8.880247 +/-1%	NA
6163302322366879832912800	6.16E22	8.670068 +/-1%	8.671369
173641788002704982030.00	1.73E20	8.591244 +/-1%	8.594962
89782960779401678872.95	8.97E19	8.641200 +/-1%	8.635581
6668512732743895218.00	6.66E18	8.571463 +/-1%	8.561422
2811690412751902758.90	2.81E18	8.526276 +/-1%	8.516395
927083404338712008.00	9.27E17	8.380986 +/-1%	8.378217
472353376312336470.00	4.72E17	8.096237 +/-1%	8.094728
45698266087910604.00	4.56E16	8.356962 +/-1%	8.361277
6107360564274685.00	6.10E15	8.144217 +/-1%	8.155518
3897826077577121.00	3.89E15	7.982160 +/-1%	7.979788
3758358676276705.00	3.75E15	7.924430 +/-1%	7.932748
2641348786316487.00	2.64E15	7.898412 +/-1%	7.910583
486153633057302.95	4.86E14	7.923999 +/-1%	7.864354
234578622590902.00	2.34E14	7.743821 +/-1%	7.723047
126430847573809.95	1.26E14	7.608452 +/-1%	7.598640
95913486256898.95	9.59E13	7.568890 +/-1%	7.581133
14223012917203.00	1.42E13	7.627279 +/-1%	7.611612
4779391590262.05	4.77E12	7.401985 +/-1%	7.395491
3556019595338.05	3.55E12	7.369693 +/-1%	7.367862
594979079066.05	5.94E11	7.429537 +/-1%	7.303338
249284579994.95	2.49E11	7.047962 +/-1%	7.023718
23065530804.95	2.30E10	6.974546 +/-1%	6.932647
1387123310.00	1.38E9	6.675363 +/-1%	6.967035
102805259.025	1.02E8	6.136616 +/-1%	6.370049

Table 2 presents a comparison of the preliminary estimates using equation (12) and the 3+ decimal place accuracy of equation (16) estimates for absolute value extreme Riemann Siegel Z function peaks $|Z_\zeta(1 + I \cdot t)|$ along the $s = 1 + I \cdot t$ line. The peak locations are nearby the peak locations reported by [19] on the critical line. Typically, given the results are from a fourier analysis grid with input spectrum spacing $\Delta t = 0.05$ the peak locations along the $s = 1 + I \cdot t$ line are only $\{-0.05, 0, 0.05\}$ different from the locations reported in [19] observed along the critical line.

Figure 5 below represents the same data as figure 3 but displayed using a |Z| versus log(log(t)) scatterplot. Qualitatively, from figure 5 it can be seen that

- the |Z| versus log(log(t)) plot of the $s = 1 + I \cdot t$ line extreme peak magnitude growth is slowing down for the highest t values in the interval shown providing some empirical evidence that the extreme peak height growth rate on the $s = 1 + I \cdot t$ line is < log(log(t)) for the interval 1e6 < t < 1e40.

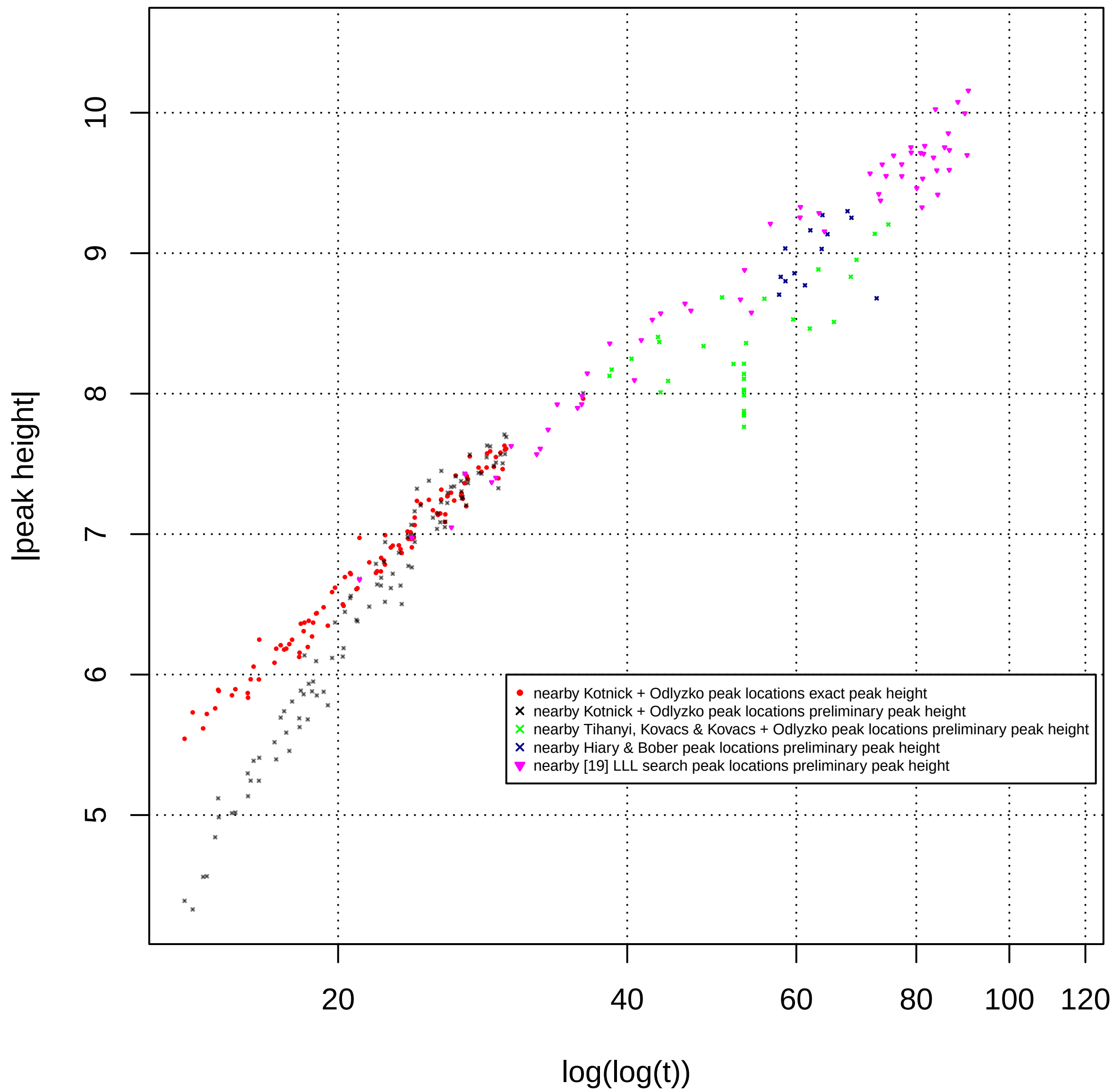

Figure 5. Comparison using |Z| versus log(log(t)) scatterplot of (i) good precision extreme Riemann Siegel Z function peak heights on the $s = 1 + I \cdot t$ line between 1e6 < t < 1e13 (red solid circles) equation (16) and (ii) preliminary extreme Riemann Siegel Z function peak heights between 1e6 < t < 1e40 based on spectral filtered heavily truncated Euler products based approximation equation (12)

$\mathbf{s = 1.5 + I \cdot t}$ line:

Using the above spectral filtering algorithm and equation (8), Figure 6 presents the |Z| versus log(t) growth of

(i) ~6 decimal place precision estimates of peak heights of extreme Riemann Siegel Z function belonging to the locations nearby reported by Kotnick [11] (solid red circles) for t < 1e13 based on equation (16),

(ii) preliminary estimates of peak heights of extreme Riemann Siegel Z function belonging to the locations nearby reported by (a) Kotnick [11] (black crosses), (b) Hiary [13], Hiary and Bober [14] (blue crosses), (c) Tihanyi et al [16-18] (green crosses) and (d) [19] (magenta triangles) using spectrally filtered heavily truncated Euler Product calculations based on equation (12),

(iii) ~6 decimal place precision estimate of peak height of a **large but not extreme** Riemann Siegel Z function peak |Z|=2.4233872854839288869837694110748369725756299326622 at $s = 1.5 + I \cdot 39246764589894309155251169284104.100622$ confirming the preliminary spectral filtering estimate from equation (12) (blue cross) nearby the reported $t = 39246764589894309155251169284104.050622$ location (black circle) which had a extreme Riemann Siegel Z function peak on the critical line reported by Hiary [13], Hiary and Bober [14], and

(iv) ~6 decimal place precision estimates of peak heights of likely extreme Riemann Siegel Z function peaks nearby $s = 1.5 + I \cdot 13656810570306744821033044688913896455457410 53$ and $s = 1.5 + I \cdot 26140941926832814423449798363364959153292041 7313$ (black circles).

Qualitatively, from figure 6 it can be seen that

- equation (12) estimates of extreme peak height (black crosses) are initially lower than the true peak height for $t < 10^9$ (red solid circles) but eventually reach reasonable agreement with the good precision equation (8) estimates between $10^9 < t < 10^{13}$.
- as mentioned there is good agreement between the preliminary estimate and equation (8) estimates for $s = 1.5 + I \cdot 39246764589894309155251169284104.100622$

Quantitatively, figure 7 uses the same extreme peak height data from figure 6 but presents the ratios |equation (12)|/|equation (8)| for peaks (i) nearby Kotnick [11], Tihanyi [16-18] and Hiary, Bober locations [13,14] (black crosses) and (ii) nearby [19] locations (magenta circles) for t < 1e21.

Hence in figure 7 it can be seen that spectrally filtered heavily truncated estimates using equation (12)

- underestimates the extreme peak height for low t for the $s = 1.5 + I \cdot t$ line
- there is a slight overshoot the extreme peak height estimate and then evidence of recovery to ratio=1 beyond 1e12. The span of available true peak height data 1e5 < t < 1e40 shows the approximations based on spectrally filtered equation (12) are reasonably converging to ratio=1 in the region 1e12 < t < 1e40 but it would good to increase the investigated region as it may become necessary to tweak the spectrally filtered equation (12) calculations by increasing Pari-GP [26] precision beyond 58 digits and/or increase the number of grid points and/or change grid spacing to maintain accuracy as $t \to \infty$.
- the spread of the ratio values for equation (14) in the region 1e10 < t < 1e21 is roughly (0.999,1.001) so a nominal accuracy performance for spectrally filtered equation (12) estimates is set at ±0.1% along the $s = 1.5 + I \cdot t$ line for 1e12 < t < 1e40.
- comparing to figure 2 of [19] and figures 2 and 4 of this paper, the accuracy of the preliminary estimates along the $s = 1.5 + I \cdot t$ line is much better ±0.1% than observed along the critical line ±10% and the Re(s)=0,1 lines ±1% which can be explained by the lower contribution of the higher frequency components of the fourier transform of $|Z_\zeta(s)|$ when Re(s) is far away from 0.5 [25].

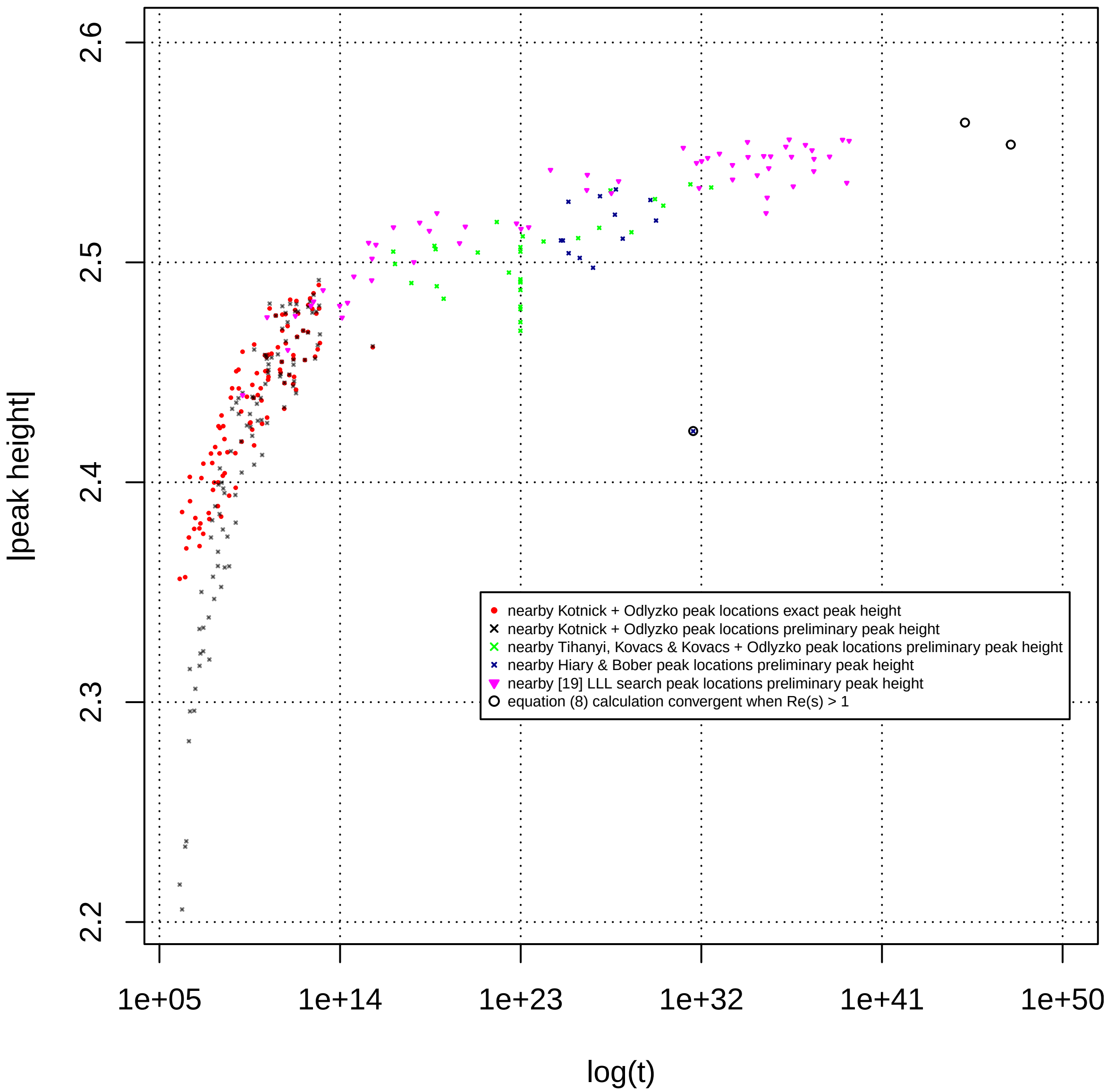

Figure 6. Comparison using log|Z| versus log(t) scatterplot of (i) good precision extreme Riemann Siegel Z function peak heights on the $s = 1.5 + I \cdot t$ line between 1e6 < t < 1e13 (red solid circles) equation (8) since Re(s) > 1, (ii) preliminary extreme Riemann Siegel Z function peak heights between 1e6 < t < 1e40 based on spectral filtered heavily truncated Euler products based approximation equation (12) and (iii) two likely extreme Riemann Siegel Z function peak heights nearby 1.36e45 and 2.61e47 (black circles).

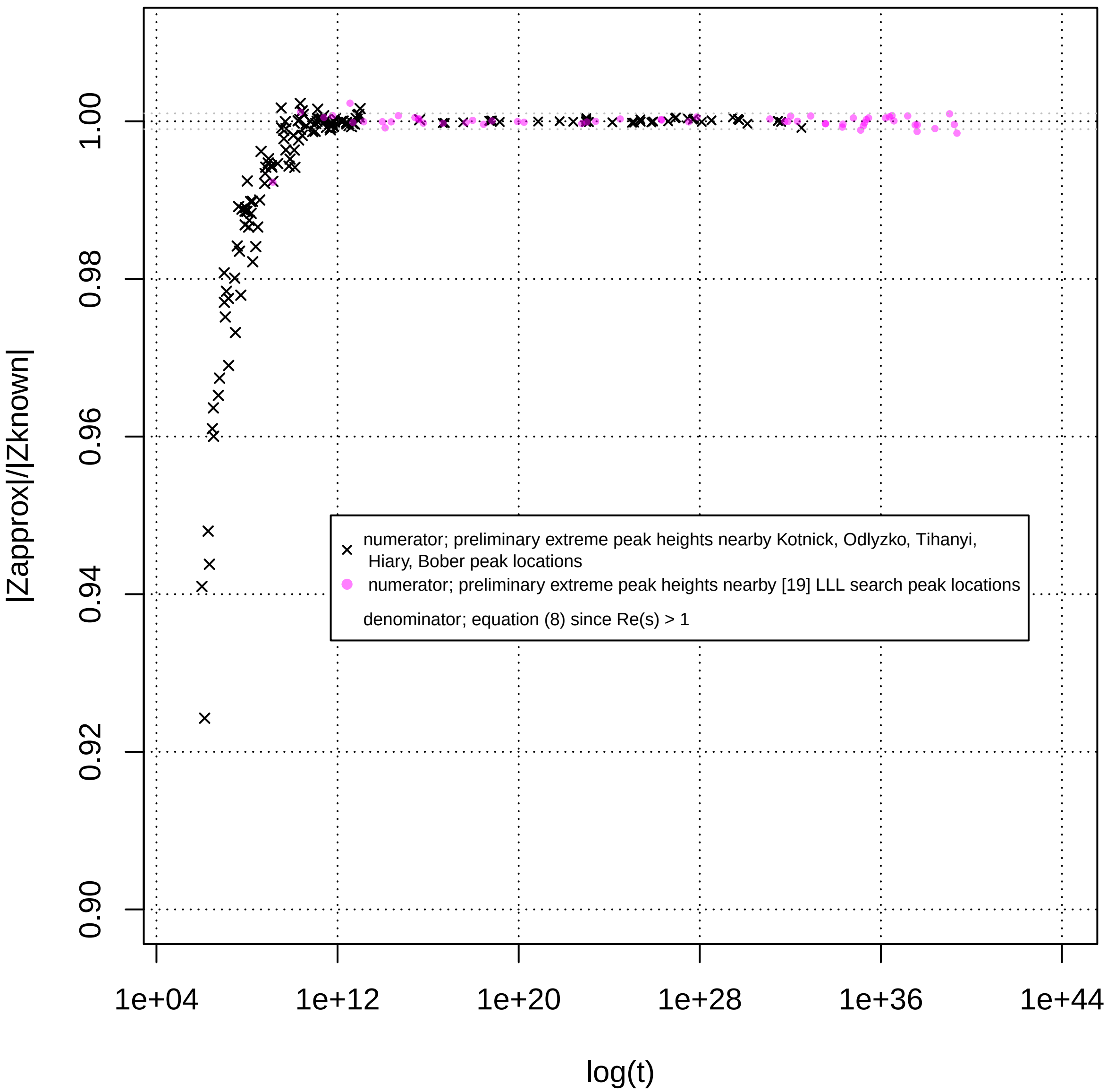

Figure 7. Ratio of spectrally filtered preliminary approximation equation (12) estimates compared to convergent equation (8) calculations on the $s = 1.5 + I \cdot t$ line between 1e6 < t < 1e40.

Table 3: Grid search location $\Delta t = 0.05$ (column 1) and preliminary height estimate of additional large Riemann Zeta peaks (column 3) using an approximate spectrally filtered Euler Product calculation employing the heavily reduced threshold $N_{\frac{\sqrt{N_1}}{2}} = \frac{1}{2} \cdot \left\lfloor \left(\frac{t}{2\pi}\right)^{0.25} \right\rfloor$. The heavy truncation in the EP calculations results in loss of the precision required for identifying Riemann Zeta zeroes but remains sufficient for preliminary estimates of (~+/- 0.1% accuracy) large peak heights on the critical line for $1e10 < t < 1e40$. The righthand column (if available) presents the |Z| value obtained using first order Riemann Siegel formula with 128 endpoint tapering calculations.

nearest grid t co-ordinate of peak	|t|	preliminary peak magnitude	|zeta(1.5+I*t)|
2302202919833091938191454510490853528294.00	2.30E39	2.555272 +/-0.1%	2.559122
1756509788486806518440112005496184630402.00	1.75E39	2.536182 +/-0.1%	2.537221
1083530769932923537813715159168803540949.00	1.08E39	2.555757 +/-0.1%	2.553335
244718139402890211698201316945603869614.95	2.44E38	2.548114 +/-0.1%	2.550482
40980891527423944866379351595135084705.00	4.09E37	2.547051 +/-0.1%	2.548184
39985965418841204149359159903547304319.00	3.99E37	2.541468 +/-0.1%	2.544745
32701888308714838788593397151662529169.00	3.27E37	2.550996 +/-0.1%	2.552184
15305431401376096542184352490753570342.00	1.53E37	2.553415 +/-0.1%	2.551681
3789770975227925929913752596153535659.00	3.78E36	2.534556 +/-0.1%	2.534435

nearest grid t co-ordinate of peak	|t|	preliminary peak magnitude	|zeta(1.5+I*t)|
3106923501574575498217324013775898531.00	3.10E36	2.548008 +/-0.1%	2.546155
2383973557258554840502898227029004956.00	2.38E36	2.555889 +/-0.1%	2.554430
1613337044611093444653528186136557938.00	1.61E36	2.552590 +/-0.1%	2.551509
284101325235663073414512322415836351.95	2.84E35	2.548209 +/-0.1%	2.547121
228437899979252082766848904790108956.00	2.28E35	2.542819 +/-0.1%	2.542288
190153064537019900206349886977800740.00	1.90E35	2.529448 +/-0.1%	2.529951
167617402698692190502094182200775332.00	1.67E35	2.522413 +/-0.1%	2.523886
128150196512905705499425650363369432.00	1.28E35	2.548348 +/-0.1%	2.551216
60758714763756271032716974136770397.00	6.07E34	2.539624 +/-0.1%	2.538547
21230740937947194961712185865535245.00	2.12E34	2.547919 +/-0.1%	2.548833
19885272645274159978060436859288428.00	1.98E34	2.554744 +/-0.1%	2.556620
3598696183652727889399753988496854.95	3.59E33	2.537662 +/-0.1%	2.538440
3546129586355636707276899029799461.00	3.54E33	2.544270 +/-0.1%	2.545005
799009489684392745192175404403618.00	7.99E32	2.549435 +/-0.1%	2.547709
207410624279318332567566187765800.95	2.07E32	2.547471 +/-0.1%	2.547459
103421228622988210350418104528308.00	1.03E32	2.546034 +/-0.1%	2.544361
77636197551578282885077537989878.00	7.76E31	2.533799 +/-0.1%	2.533855
57402689537187327152922167612334.00	5.74E31	2.545204 +/-0.1%	2.545234
12563916677893023747500097076528.00	1.25E31	2.552095 +/-0.1%	2.551349
7536416471667947828910808698.00	7.53E27	2.536914 +/-0.1%	2.535484
3286305986924467937604770718.00	3.28E27	2.531476 +/-0.1%	2.531652
207977319495059838955685936.00	2.07E26	2.539807 +/-0.1%	2.539410
195379681268348910290306782.00	1.95E26	2.532890 +/-0.1%	2.532328
3047726497970993730122699.00	3.04E24	2.542064 +/-0.1%	2.541322
249774893717336617897828.00	2.49E23	2.515978 +/-0.1%	2.516000
103594925224774818957500.00	1.03E23	2.515232 +/-0.1%	2.515450
61633023223668798329128.00	6.16E22	2.517707 +/-0.1%	2.518445
173641788002704982030.00	1.73E20	2.516260 +/-0.1%	2.516562
89782960779401678873.00	8.97E19	2.508738 +/-0.1%	2.508829
6668512732743895218.00	6.66E18	2.522367 +/-0.1%	2.522254
2811690412751902758.90	2.81E18	2.514326 +/-0.1%	2.515289
927083404338712008.00	9.27E17	2.518068 +/-0.1%	2.517770
472353376312336470.00	4.72E17	2.500111 +/-0.1%	2.500568
45698266087910604.00	4.56E16	2.515949 +/-0.1%	2.516590
6107360564274685.00	6.10E15	2.508011 +/-0.1%	2.508592
3897826077577121.00	3.89E15	2.501695 +/-0.1%	2.500889
3758358676276705.00	3.75E15	2.491838 +/-0.1%	2.491342
2641348786316487.00	2.64E15	2.508899 +/-0.1%	2.507711
486153633057302.95	4.86E14	2.493563 +/-0.1%	2.491783
234578622590902.00	2.34E14	2.481613 +/-0.1%	2.481785
126430847573810.00	1.26E14	2.474888 +/-0.1%	2.476963
95913486256899.00	9.59E13	2.480231 +/-0.1%	2.480368
14223012917202.95	1.42E13	2.487345 +/-0.1%	2.487493
4779391590262.05	4.77E12	2.482215 +/-0.1%	2.482581
3556019595338.00	3.55E12	2.480275 +/-0.1%	2.474587
594979079066.05	5.94E11	2.475629 +/-0.1%	2.473842
249284579994.95	2.49E11	2.460146 +/-0.1%	2.458958
23065530805.00	2.30E10	2.475044 +/-0.1%	2.472195
1387123310.00	1.38E9	2.439626 +/-0.1%	2.458610
102805259.025	1.02E8	2.406315 +/-0.1%	2.424678

Table 3 presents a comparison of the preliminary estimates using equation (12) and the 6+ decimal place accuracy of equation (8) estimates for absolute value extreme Riemann Siegel Z function peaks $|Z_\zeta(1.5 + I \cdot t)|$ along the $s = 1.5 + I \cdot t$ line. The peak locations are nearby the peak locations reported by [19] on the critical line. Typically, given the results are from a fourier analysis grid with input spectrum spacing $\Delta t = 0.05$ the peak locations along the $s = 1.5 + I \cdot t$ line are only $\{-0.05, 0, 0.05\}$ different from the locations reported in [19] observed along the critical line.

Conclusions

In the region 1e10 < t < 1e40 the use of spectrally filtered heavily truncated partial Euler Products approximations of the Riemann Siegel Z function shows potential for preliminary estimation of extreme peak height growth behaviour across the complex plane. The use of heavy truncation and short spectrum lengths for fourier analysis appears applicable to extreme peak height estimation and this approach may provide a useful tool for pre-screening LLL algorithm [35] search results prior to high precision calculations.

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Appendix - Existing methods for calculating Riemann Siegel Z function peak height

Equation (9) itself is calculated for the Riemann Siegel Z function at the identified peak position either using one of four methods depending on the value of the t-coordinate and whether Re(s) > 1.

At the lowest t values

$$|Z_\zeta(s)| = \left| e^{I \cdot imag(\theta_{ext}(s))} \zeta(s) \right| \tag{14}$$

using the intrinsic Pari-GP function [26] employing the Euler-Maclaurin approximation.

For $p \cdot \pi < t < p^2 \cdot 2 \cdot \pi$ by tapered truncation at the second quiescent region using the Riemann Zeta Dirichlet Series $\lfloor \frac{t}{\pi} \rfloor$ approximation

$$\left| e^{I \cdot imag(\theta_{ext}(s))} \zeta(s) \right| \approx \left| e^{I \cdot imag(\theta_{ext}(s))} \cdot \left[\sum_{k=1}^{(\lfloor \frac{t}{\pi} \rfloor - p)} \left(\frac{1}{k^{(s)}} \right) + \sum_{i=(-p+1)}^{p} \frac{\frac{1}{2^{2p}} \left(2^{2p} - \sum_{k=1}^{i+p} \binom{2p}{2p-k} \right)}{\left((\lfloor \frac{t}{\pi} \rfloor + i)^{(s)} \right)} \right] \right|, \qquad \Im(s) \to \infty \tag{15}$$

giving multidecimal agreement using 2p=128 endpoint tapering with the Euler-Maclaurin approximation $t > p \cdot \pi$.

At $t > p^2 \cdot 2 \cdot \pi$ by tapered truncation of the first order Riemann Siegel formula terms [29-31] at the first quiescent region using the Riemann Zeta Dirichlet Series $\left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor$ approximation

$$\begin{aligned} \left| e^{I \cdot imag(\theta_{ext}(s))} \zeta(s) \right| \approx \Bigg| e^{I \cdot imag(\theta_{ext}(s))} \cdot \Bigg[& \sum_{k=1}^{(\lfloor \sqrt{\frac{t}{2\pi}} \rfloor - p)} \left(\frac{1}{k^{(s)}} \right) + \sum_{i=(-p+1)}^{p} \frac{\frac{1}{2^{2p}} \left(2^{2p} - \sum_{k=1}^{i+p} \binom{2p}{2p-k} \right)}{\left((\lfloor \sqrt{\frac{t}{2\pi}} \rfloor + i)^{(s)} \right)} \\ & + \chi(s) \cdot \left(\sum_{k=1}^{(\lfloor \sqrt{\frac{t}{2\pi}} \rfloor - p)} \left(\frac{1}{k^{(1-(s))}} \right) + \sum_{i=(-p+1)}^{p} \frac{\frac{1}{2^{2p}} \left(2^{2p} - \sum_{k=1}^{i+p} \binom{2p}{2p-k} \right)}{\left((\lfloor \sqrt{\frac{t}{2\pi}} \rfloor + i)^{(1-(s))} \right)} \right) \Bigg] \Bigg| \\ & , \qquad \Im(s) \to \infty \end{aligned} \tag{16}$$

giving 3+ decimal place accuracy (for 2p=128 endpoint tapering) and improving as t increases, where 2p is the number of end taper weighted points present in the second term of the equation and (iii) $\binom{2p}{2p-k}$ are the binomial coefficients. Away from the real axis these tapered Dirichlet Series sums provides an excellent approximation of its Riemann Zeta and Riemann Siegel Z function analogues.

At the highest t values t > 1e21, the only available confirmed high precision extreme Riemann Siegel Z function heights used in this paper for $Re(s) \neq 0.5$ are those obtained via equation (8) when Re(s)=1.5.